

JadeDragon: Optical Music Recognition for Kunqu Opera Gongche Notation

End-to-end transcription of handwritten gongche notated 工尺譜 scores into MusicXML and audio

Kelly Yang · Jiacheng Dai · Andrew Park

Brown University · CSCI 1430 Computer Vision · 2026

01 Motivation

Kunqu Opera (昆曲) — six centuries old, UNESCO Masterpiece of the Oral and Intangible Heritage of Humanity (2001) — preserves its vocal repertoire in **gongche notation** (工尺譜). Most surviving scores are handwritten, calligraphic manuscripts that only specialists can read.

Western OMR systems (Audiveris, Oemer) ignore non-Western pitch systems and lyric alignment, so this corpus is **decaying with its readers**.

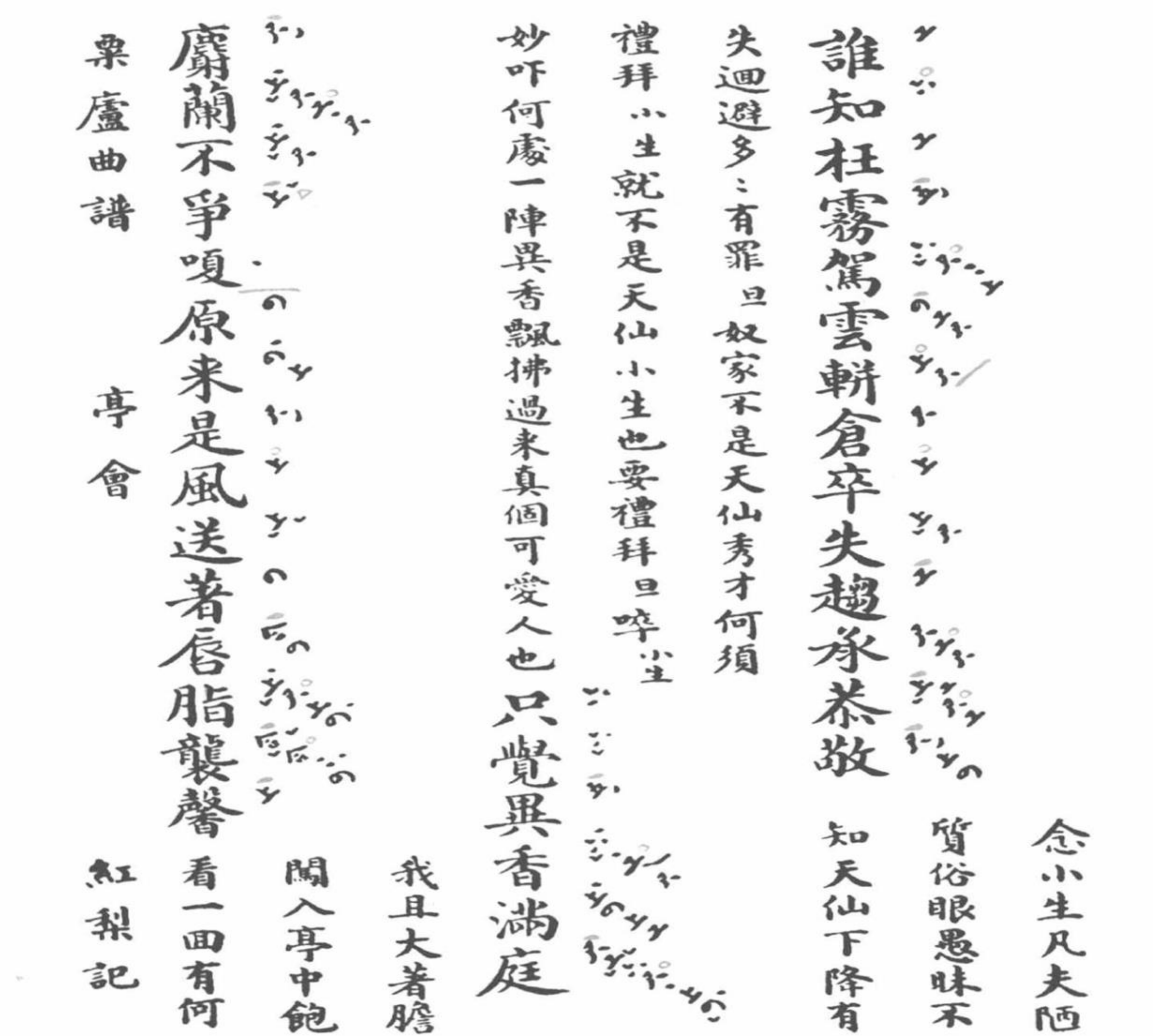


Fig 1. Lili Qupu (栗廬曲譜) — calligraphic kaishu lyrics with gongche pitches in the upper-right of each lyric character. Vertical columns read top-to-bottom, right-to-left.

02 Problem

Given a scan of a Kunqu gongche page, produce a structured score (MusicXML + MIDI) that captures:

- Detection of 28 pitch classes — gongche characters with octave-modified forms (𠂇 上, 上,, etc.)
- Recognition of vertical CJK lyrics in calligraphic kaishu — out-of-distribution for printed-text OCR engines
- Cross-modal alignment: each lyric carries a melisma of one or more pitches in the column to its right
- Inference of key signature (調) and transposition to concert pitch for playback

03 Goal and the Data Set

- Single page-in / score-out pipeline that runs on real Lili Qupu (栗廬曲譜) scans without per-page tuning
- Match or exceed the LGRC2024 paper baseline on pitch detection (74.3% mAP50) — achieved ~81% mAP50, +6.7 over baseline
- Produce audibly playable MIDI a non-expert listener can hear — useful, not just publishable
- Re-frame Kunqu OMR as an AMNLT-style aligned-music-and-lyrics task to enable comparison with non-Western OMR work
- **Datasets** — Pitch (LGRC2024): 2,134 gongche pages (1,496/638 split), 28 classes — 7 key signatures + 21 pitch symbols (C3–B5). Lyric (ours): 51 hand-labeled Lili Qupu (栗廬曲譜) pages (40/11 split, 2029 boxes).

04 What is Gongche Notation?

A character-based pitch notation dating to the Tang dynasty, written in vertical columns, read right to left. Pitches use **movable-do solfège** — the gongche character maps to a scale degree, and the key signature 調 sets the tonal center.

Gongche	上	尺	工	凡	六	五	乙
Solfège	do	re	mi	fa	sol	la	ti
Degree	1	2	3	4	5	6	7

𠂇 prefix → octave up, 𠂇 suffix → octave down, 六字調 = F major

05 Key Insight

Pitches → object detection, not OCR

The gongche symbol set is closed (28 classes - 7 key signatures, 21 distinct notes) and spatial — 上 vs 𠂇 上 vs 上, are different pitches. OCR returns text strings; detection returns boxes with class labels we can geometrically align with lyrics.

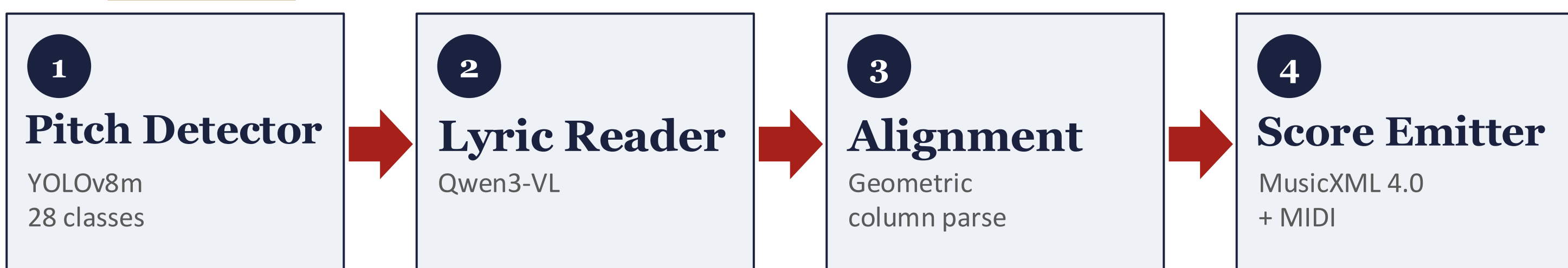
Calligraphic CJK → detection-first pipeline

OCR engines fail on Lili Qupu's brushwork. We hand-label 51 pages via VIA + active learning and train a single-class YOLOv8m reaching 99.3% mAP50. Per-box Qwen3-VL recognition achieves ~95–98% accuracy on detected crops.

Alignment is geometric, not learned

Kunqu reading order is fixed: top-to-bottom within columns, right-to-left across columns; gongche sits in the upper-right of each lyric. A center-y heuristic suffices — no learned alignment needed.

06 Method



Stage 1 — Pitch Detection. YOLOv8m (Ultralytics) trained 110 epochs on the LGRC2024 dataset (1,496 train / 638 val pages) on a Brown Oscar A40. We disabled rotation/flip augmentation — Kunqu reading order is fixed and rotated pages are out-of-distribution at test time. This single change drove +6.7 mAP50 over the paper baseline.

Stage 2 — Lyric Reading. 51 hand-labeled pages (40/11 train/val, 2029 boxes) via VIA + active-learning picker. Single-class YOLOv8m achieves 99.3% mAP50. Per-box Qwen3-VL calls recognize each crop; page 150: 45/54 chars found, 43 correctly identified (80%).

Stage 3 — Alignment. Within each column, every detected pitch is assigned to the lyric whose center-y is just below the pitch's center-y, producing (lyric, [pitches]) tuples. Cross-column ordering is right-to-left.

Stage 4 — Score Emission. Each pitch character maps to a C-major reference (上 = C4) and is transposed by the detected key signature. MusicXML 4.0 attaches lyric syllables as melismas; FluidSynth renders MIDI to audio.

Limitations & Future Work

- Rhythm reconstruction.** Detect ban (。) and yan (、) marks to recover note durations and bar boundaries — currently the largest correctness gap. Would require new object classes and labels.
- Generalization.** Trained on Lili Qupu only; untested on Yuzhu, Yizi, Nanyin, and the 32-volume Jicheng Qupu corpus.
- Synthetic pretraining.** Gongche encoding rules are deterministic; lyric corpora are abundant — procedural rendering enables unlimited paired (image, MusicXML) data.

07 Results

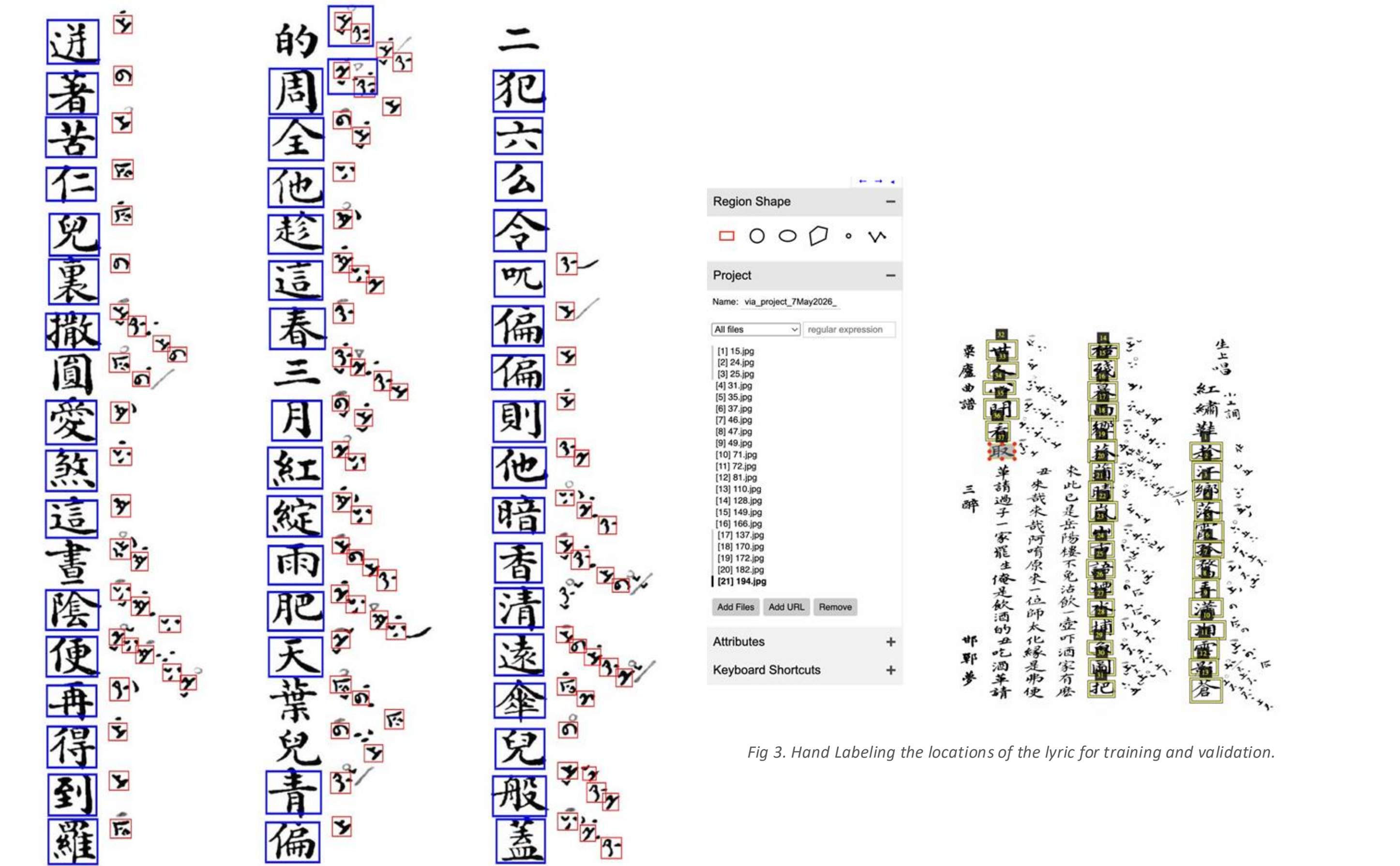
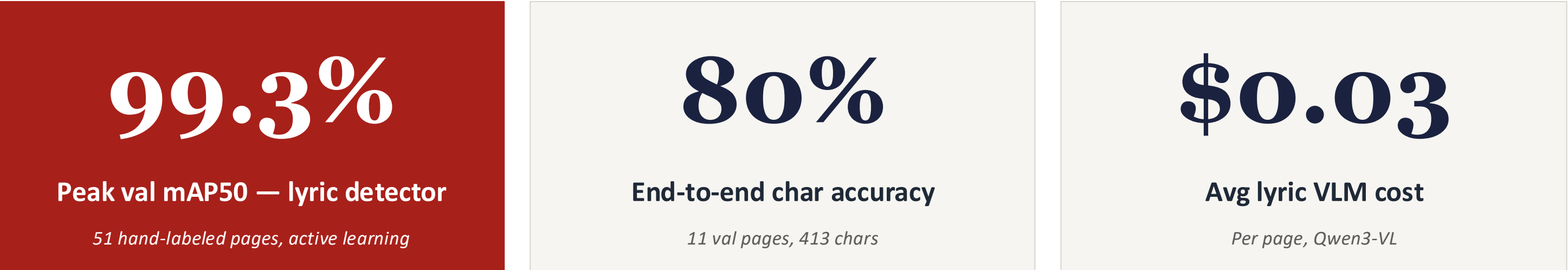


Fig 2. End-to-end output on val page 150. Blue boxes: lyric character bbox detections. Red boxes: 28-class pitch detector predictions.

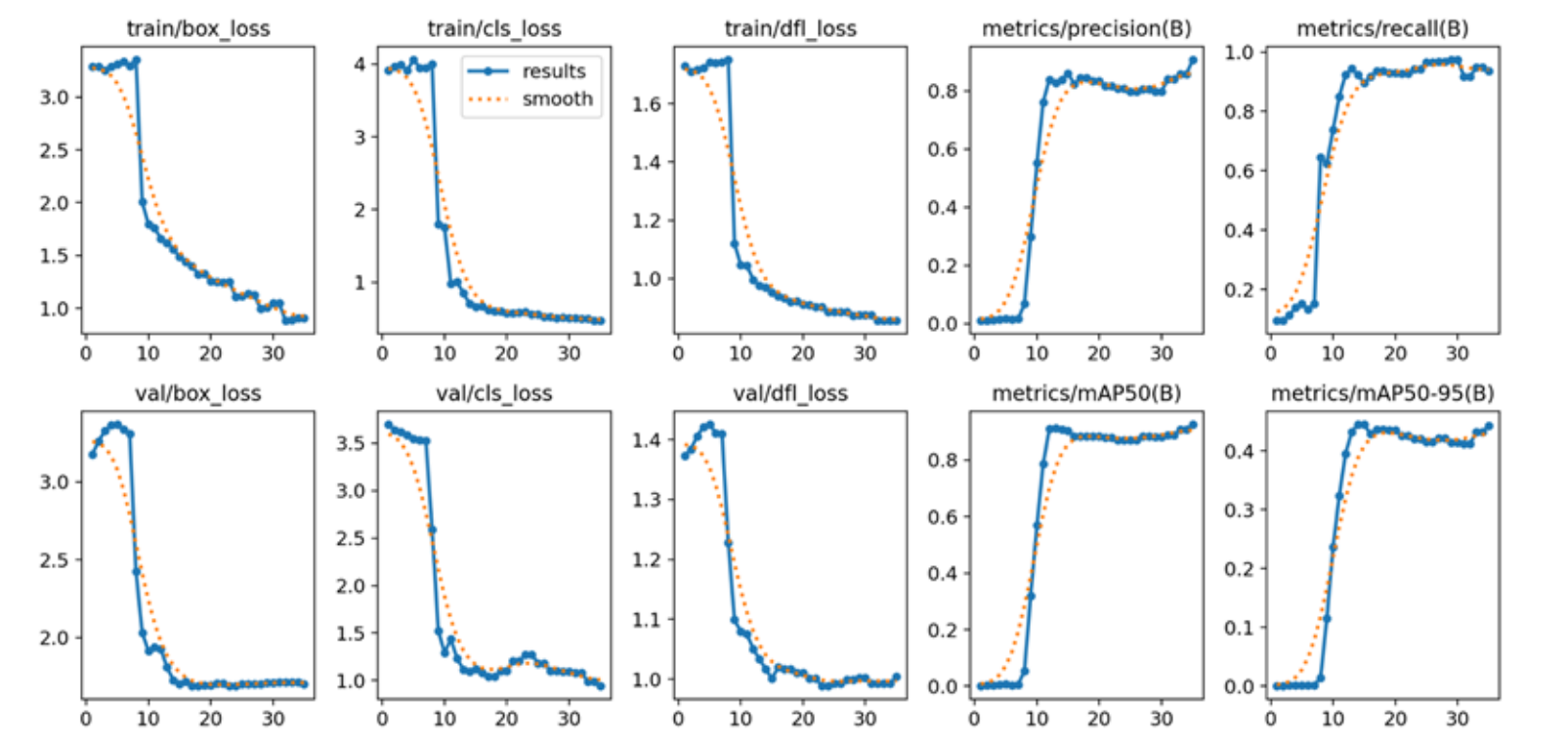


Fig 5. Training dynamics on Brown Oscar A40.

- 尋夢 (Xun Meng), page 150: 45/54 lyric chars found, 43 correctly recognized by Qwen3-VL (80% end-to-end). 六字調 → F major, audibly playable MIDI.
- Disabling rotation/flip augmentation — counter to LGRC2024 — yielded +6.7 mAP50; rotated pages are not in the real test distribution.
- Failure modes: ~7% adjacent-pitch confusion; rare classes (A3, A4, D5) below 0.5 AP — a class-imbalance dataset issue, not a model issue.

08 References & Acknowledgements

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- [2] Chen, G. (2014). An optical music recognition system for traditional Chinese Kunqu Opera scores. *EURASIP J. Audio Speech Music Process.*
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- [4] Jocher, G. et al. (2023). Ultralytics YOLOv8.
- [5] Qwen3-VL technical report — Alibaba (2024).
- [6] MusicXML 4.0 specification, W3C Music Notation CG.

Acknowledgements. Brown CCV (Oscar) for A40 compute · CSCI 1430 staff · Lili Qupu transcribers and Kunqu performers whose labor made this corpus exist.